

Task is to build 8-bit Microcontroller: 04/04/2024

A microcontroller

A microcontroller is a type of integrated circuit that is represented as a small personal computer (PC) without operating systems (OS) intended to control specific functions of larger components. It consists of a single chip with an input/output (I/O) peripheral, memory, and a central processing unit (CPU).

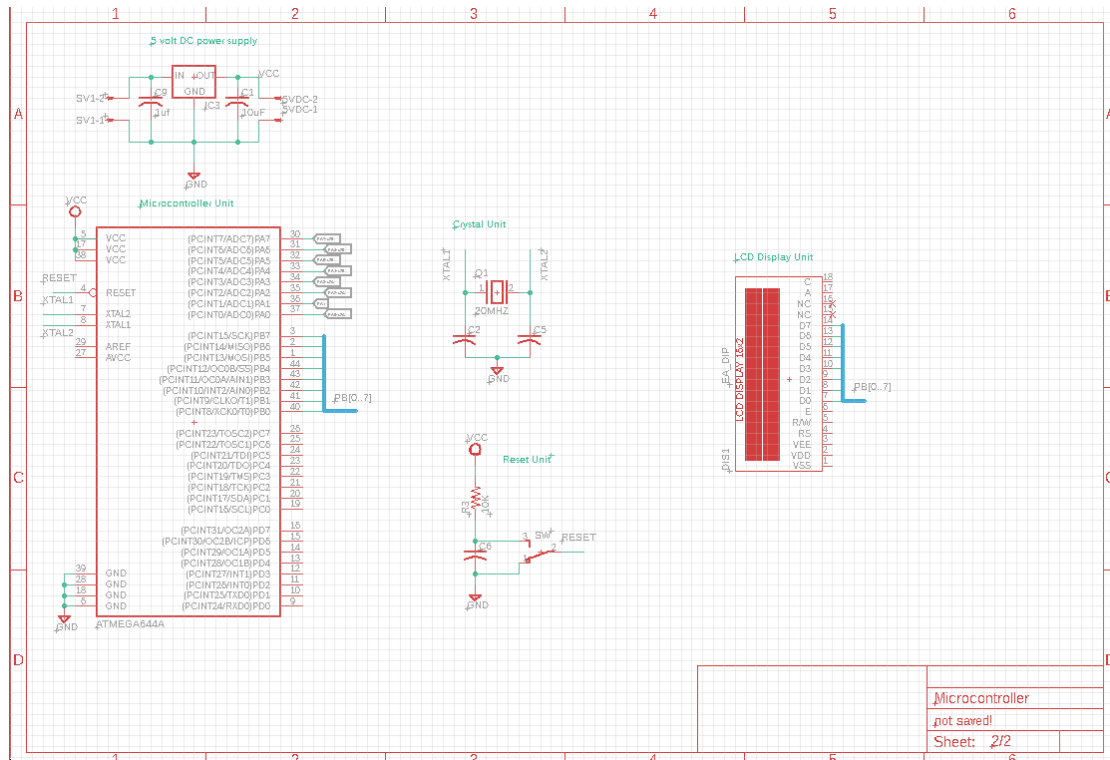
Difference Between Microprocessor and Microcontroller

Microprocessor	Microcontroller
Used in personal computers.	Used in embedded systems.
Complex and expensive due to a large number of instructions to process.	Simple and inexpensive due to a smaller number of instructions to process.
Large in size, as the memory and I/O are connected externally	Small in size as the memory and I/O are present together
RAM, ROM, I/O units, and other peripherals are not embedded on a single chip.	RAM, ROM, CPU and other peripherals are embedded on a single chip.

- **Global connector/data nets:** There are two types of connection the first one global connection can be used for more than one sheet such as GND, VCC, VDD, VEE, while the other types are called label off page (data nets) are used in the same sheet only.

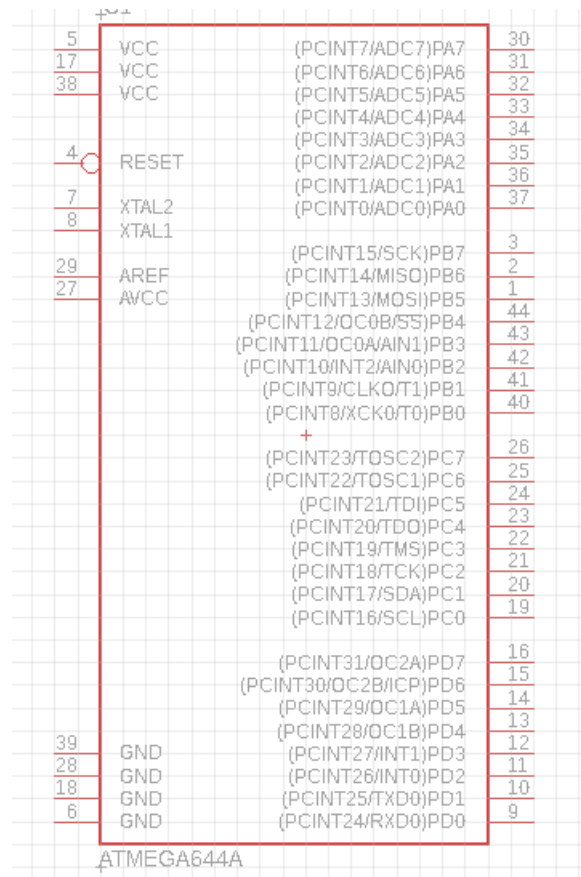
ATmega328p" used in the Arduino Uno.

Microcontroller: ATMEGA644A



1. Microcontroller: ATMEGA644A

<https://docs.rs-online.com/c2d4/0900766b8166ecd0.pdf>



Name	Managed Folder	Description	Popularity
atmel	Eagle Pcb	AVR Devices	
AT8*CS1S...		Single-Chip Flash Mi...	
AT45DB1...		16-megabit 2.5V or...	
AT45DB6...		Flash-Memory 64-M...	
AT89C51-*		8-bit Microcontroller...	
AT89C517*		8-bit Microcontroller...	
AT89C513*		8-bit Flash Microcon...	
AT89C5132		USB Microcontroller...	
AT89S51*P		8-bit Microcontroller...	
AT89S51-*		8-bit Microcontroller...	
AT91SAM...		Highly integrated FL...	
AT91SAM...		Highly integrated FL...	
AT93C46...		Three-wire Automo...	
ATF16V88		High performance C...	
ATMEGA3...		8-bit Microcontroller...	
ATMEGA6...		8-bit Microcontroller...	
ATMEGA6...		8-bit Microcontroller...	
ATMEGA6...		8-bit Microcontroller...	
MEGA32		MICROCONTROLLER	
MEGA32-M		MICROCONTROLLER	
MEGA32-P		MICROCONTROLLER	
T87C5121		8-bit Microcontroller...	
T8C5121		8-bit Microcontroller...	
T89C51CC...		Enhanced 8-bit Micr...	
T89C51CC...		Enhanced 8-bit Micr...	
T7024		Bluetooth/ISM 2.4-...	
TINY13		8-bit AVR Microcon...	
TINY13*		8-bit AVR Microcon...	
TINY26*		8-bit AVR Microcon...	
TINY26*-M		8-bit AVR Microcon...	

5mm
0.2in

ATMEGA325P (Version 2)

8-bit Microcontroller with 32K Bytes In-System Programmable Flash

Source: <http://www.atmel.com..doc8023.pdf>

Footprint: 64M1 (Version 1)

64M1

64-pad, 9 x 9 x 1.0 mm

Lead Pitch 0.50 mm, 5.40 mm Exposed Pad, Micro Lead Frame Package

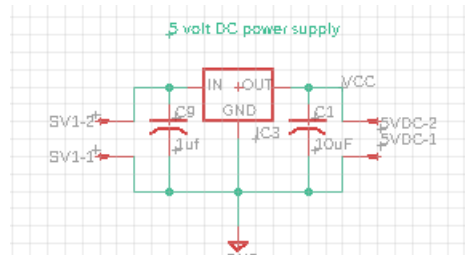
Attribute	Value
MF	
MPN	ATMEGA325PV-10AU
OC_FARNELL	1636939
OC_NEWARK	03P2025
POPULARITY	0

2. Power supply 12v to have 5 volts with two capacitors

Used v-reg 78LXX and from the circuit application in the following circuit used two capacitors.C1=10uF and C2=1uF

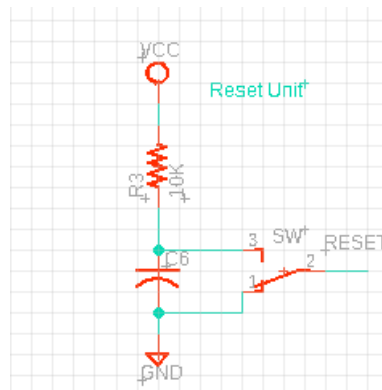
<https://reversepcb.com/78l05-voltage-regulator/>

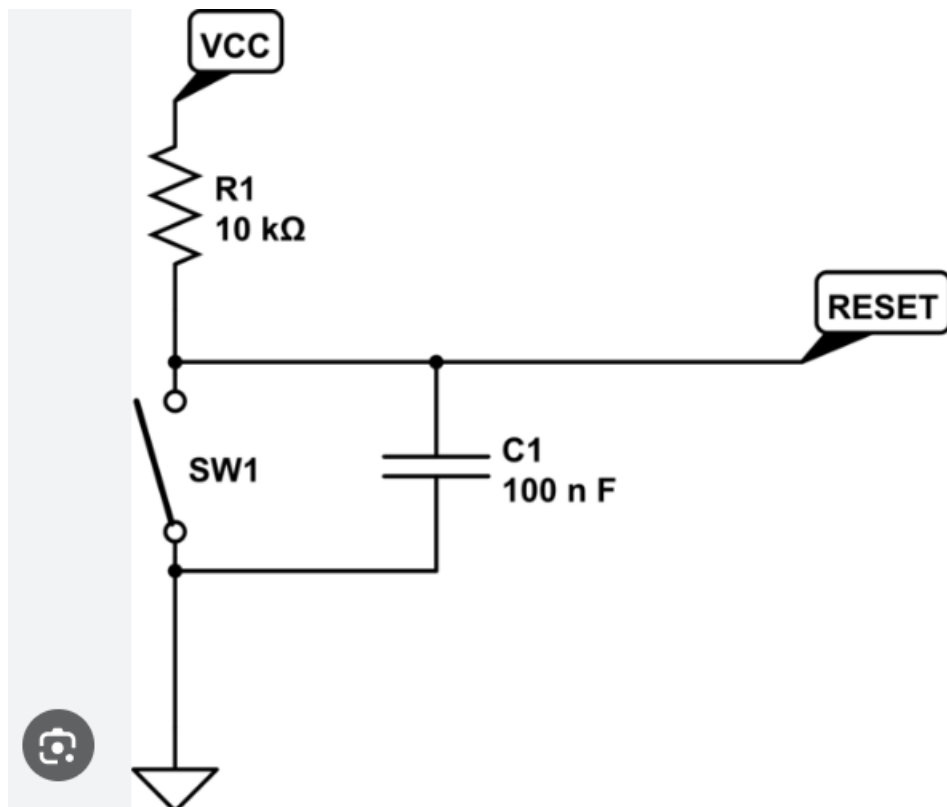
<https://www.snapeda.com/>



3. Rest switch for microcontroller:

C=100nf and R=10KOhm

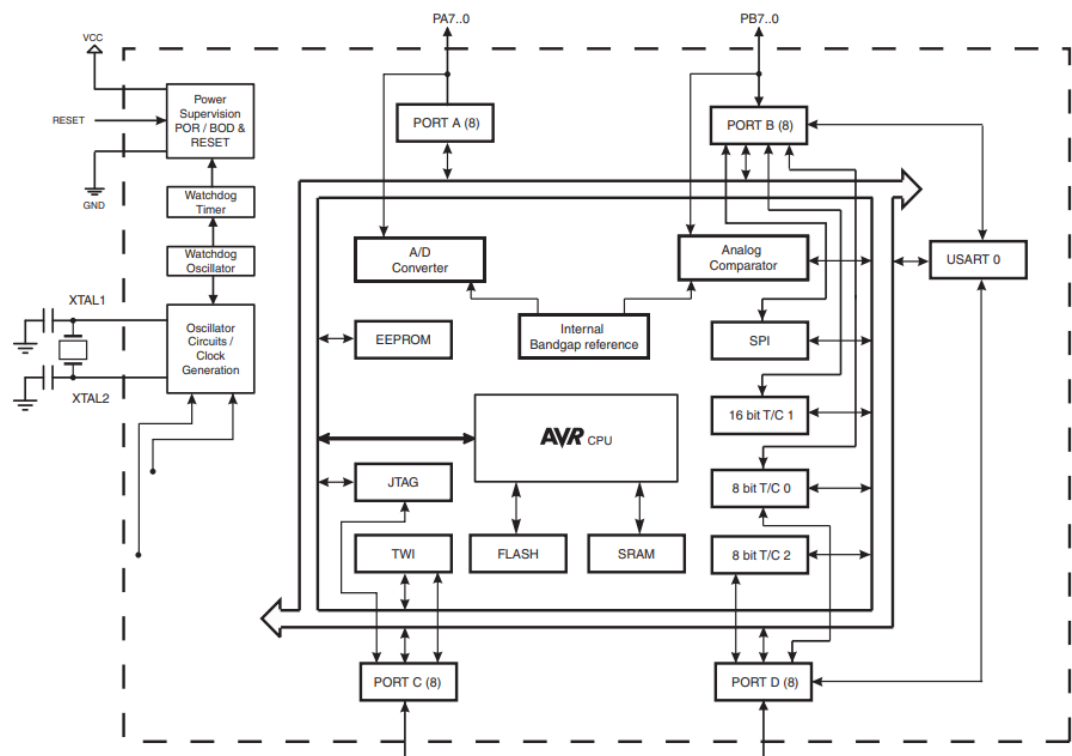




microcontroller - Use or remove reset capacitor ...

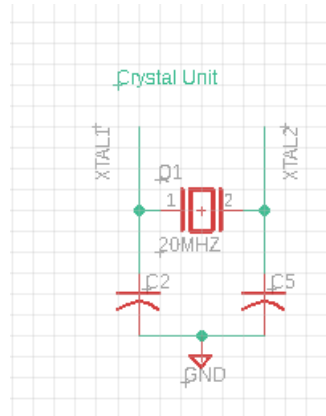
V

4. Crystal Oscillator



6.1 ATmega644

Speed (MHz) ⁽³⁾	Power Supply	Ordering Code ⁽²⁾	Package ⁽¹⁾	Operational Range
10	1.8V - 5.5V	ATmega644V-10AU	44A	Industrial (-40°C to 85°C)
		ATmega644V-10PU	40P6	
		ATmega644V-10MU	44M1	
20	2.7V - 5.5V	ATmega644-20AU	44A	Industrial (-40°C to 85°C)
		ATmega644-20PU	40P6	
		ATmega644-20MU	44M1	



5.LED 8 bits data connections

Need to connect the display unit with bus connection

